CaSB 150 — Complete Mathematics Reference

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1. Logarithm & Exponential Algebra

These identities are the foundation of everything. Memorize them and the steps to apply them.

Core Log & Exponential Identities

$$e^{\ln x} = x \quad \text{(they undo each other)} \tag{1}$$
$$\ln(e^x) = x \tag{2}$$
$$\ln(AB) = \ln A + \ln B \tag{3}$$
$$\ln(A/B) = \ln A - \ln B \tag{4}$$
$$\ln(A^r) = r \ln A \tag{5}$$
$$e^{a+b} = e^a \cdot e^b \tag{6}$$
$$e^{a-b} = e^a / e^b \tag{7}$$
$$(e^a)^b = e^{ab} \tag{8}$$

1.1 Converting $\ln N = rt + C$ to $N(t)$

This exact manipulation happens in **every** growth equation solution.

Step 1: Exponentiate both sides
(apply $e^{(\cdot)}$ to both sides):

$$e^{\ln N} = e^{rt+C}$$

Step 2: Left side simplifies by identity (1):

$$N = e^{rt+C}$$

Step 3: Right side: split exponent using identity (6):

$$N = e^{rt} \cdot e^C$$

Step 4: Rename e^C as a new constant N_0 (since $e^C > 0$ always):

$$\boxed{N(t) = N_0\, e^{rt}}$$

Step 5: Apply initial condition $N(0) = N_0$: plug in $t = 0$:

$$N(0) = N_0 e^0 = N_0 \cdot 1 = N_0 \quad \checkmark$$

Worked Example: Finding r from doubling time

If N doubles every T days (i.e. $N(T) = 2N_0$):

$$N_0 e^{rT} = 2N_0 \implies e^{rT} = 2 \implies rT = \ln 2 \implies \boxed{r = \frac{\ln 2}{T}}$$

If instead N multiplies by e every T days ($N(T) = e \cdot N_0$):

$$e^{rT} = e = e^1 \implies rT = 1 \implies \boxed{r = \frac{1}{T}}$$

1.2 Converting Discrete to Continuous

Given discrete growth $N(t+1) = \lambda N(t)$ with $\lambda = 1 + r_d$:

$$\lambda = e^{r_c} \implies r_c = \ln(\lambda) = \ln(1 + r_d)$$

Step 1: For small r_d (use $\ln(1+x) \approx x$ for $x \ll 1$):

$$r_c \approx r_d \quad \text{(discrete and continuous rates are approximately equal)}$$

Step 2: If you know r_c and want λ:

$$\lambda = e^{r_c}$$

Step 3: Effective rate from piecewise growth (rate r_1 for time t_1, rate r_2 for time t_2):

$$N(t_1 + t_2) = N_0\, e^{r_1 t_1} \cdot e^{r_2 t_2} = N_0\, e^{r_1 t_1 + r_2 t_2}$$
$$r_{\text{eff}} = \frac{r_1 t_1 + r_2 t_2}{t_1 + t_2} \quad \text{(time-weighted average)}$$

2. Derivatives — Complete Rules

Derivative Rules You Need

Rule	Formula	Example
Power rule	$\frac{d}{dx}[x^n] = nx^{n-1}$	$\frac{d}{dx}[N^3] = 3N^2$
Exponential	$\frac{d}{dx}[e^{rx}] = re^{rx}$	$\frac{d}{dt}[e^{5t}] = 5e^{5t}$
Natural log	$\frac{d}{dx}[\ln x] = \frac{1}{x}$	$\frac{d}{dN}[\ln N] = \frac{1}{N}$
Constant mult.	$\frac{d}{dx}[cf(x)] = cf'(x)$	$\frac{d}{dx}[3N^2] = 6N$
Sum/Difference	$(f \pm g)' = f' \pm g'$	—
Product rule	$(fg)' = f'g + fg'$	see below
Quotient rule	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$	see below
Chain rule	$(f(g(x)))' = f'(g(x)) \cdot g'(x)$	see below

2.1 Product Rule — Full Steps

Worked Example: $\frac{d}{dt}[N(t) \cdot P(t)]$

Step 1: Label $f = N(t)$, $g = P(t)$.
Step 2: Apply formula $(fg)' = f'g + fg'$:

$$\frac{d}{dt}[NP] = \frac{dN}{dt} \cdot P + N \cdot \frac{dP}{dt} = \dot{N}P + N\dot{P}$$

This appears when differentiating NP terms in predator-prey models.

2.2 Quotient Rule — Full Steps (Type II Functional Response)

Worked Example: $\frac{d}{dN}\left[\frac{\alpha N}{1 + \alpha h N}\right]$

This is the derivative of the Type II functional response, needed for the 2024 Jacobian.
Step 1: Label numerator $f = \alpha N$, denominator $g = 1 + \alpha h N$.
Step 2: Compute individual derivatives:

$$f' = \frac{d}{dN}[\alpha N] = \alpha, \qquad g' = \frac{d}{dN}[1 + \alpha h N] = \alpha h$$

Step 3: Apply quotient rule $\frac{f'g - fg'}{g^2}$:

$$= \frac{\alpha(1 + \alpha h N) - \alpha N \cdot \alpha h}{(1 + \alpha h N)^2}$$

Step 4: Expand the numerator:

$$= \frac{\alpha + \alpha^2 h N - \alpha^2 h N}{(1 + \alpha h N)^2} = \frac{\alpha}{(1 + \alpha h N)^2}$$

Step 5: Notice the original function was $\bar{\beta}(N) = \frac{\alpha N}{1 + \alpha h N}$, so $\bar{\beta}/N = \frac{\alpha}{1 + \alpha h N}$. Thus:

$$\boxed{\frac{d}{dN}\left[\frac{\alpha N}{1+\alpha hN}\right] = \frac{\alpha}{(1+\alpha hN)^2}}$$

2.3 Chain Rule — Full Steps

Worked Example: $\frac{d}{dt}[\ln(N(t))]$

Step 1: Outer function: $f(u) = \ln(u)$, **so** $f'(u) = 1/u$.
Step 2: Inner function: $u = N(t)$, **so** $u' = dN/dt = \dot{N}$.
Step 3: Chain rule: $\frac{d}{dt}[\ln N] = \frac{1}{N} \cdot \dot{N}$:

$$\boxed{\frac{d}{dt}[\ln N(t)] = \frac{1}{N}\frac{dN}{dt}}$$

This is the **per-capita growth rate**. If $\dot{N} = rN$, then $\frac{d}{dt}[\ln N] = r$.

Worked Example: $\frac{d}{dt}[e^{-t/\tau}]$

Step 1: Outer function: $f(u) = e^u$, **inner function:** $u = -t/\tau$.
Step 2: $f'(u) = e^u$, $u' = -1/\tau$.
Step 3: Chain rule:

$$\frac{d}{dt}[e^{-t/\tau}] = e^{-t/\tau} \cdot \left(-\frac{1}{\tau}\right) = -\frac{1}{\tau}e^{-t/\tau}$$

3. Partial Derivatives (for Jacobian Matrices)

The Rule

A **partial derivative** $\frac{\partial f}{\partial x}$ means: differentiate f with respect to x, **treating all other variables as constants**.

3.1 Step-by-Step Partial Derivative Process

Worked Example: System: $f(N, P) = rN - \bar{\beta}NP$

Step 1: Find $\partial f/\partial N$ — treat P as a constant number:

$$\frac{\partial}{\partial N}[rN - \bar{\beta}NP] = \frac{\partial}{\partial N}[rN] - \frac{\partial}{\partial N}[\bar{\beta}NP]$$

For the second term, $\bar{\beta}P$ is just a constant times N:

$$= r - \bar{\beta}P$$

Step 2: Find $\partial f/\partial P$ — treat N as a constant number:

$$\frac{\partial}{\partial P}[rN - \bar{\beta}NP] = \frac{\partial}{\partial P}[rN] - \frac{\partial}{\partial P}[\bar{\beta}NP]$$

rN has no P in it, so its partial is 0. $\bar{\beta}N$ is a constant times P:

$$= 0 - \bar{\beta}N = -\bar{\beta}N$$

Worked Example: System: $g(N, P) = \varepsilon\bar{\beta}NP - ZP$

Step 3: $\partial g/\partial N$: treat P as constant.

$$\frac{\partial g}{\partial N} = \varepsilon\bar{\beta}P - 0 = \varepsilon\bar{\beta}P$$

Step 4: $\partial g/\partial P$: treat N as constant.

$$\frac{\partial g}{\partial P} = \varepsilon\bar{\beta}N - Z$$

Worked Example: Logistic term: $f(N,P) = rN(1-N/K) - \bar{\beta}NP$

Step 5: Expand first: $f = rN - rN^2/K - \bar{\beta}NP$.
Step 6: $\partial f/\partial N$: N^2 becomes $2N$ by power rule:

$$\frac{\partial f}{\partial N} = r - \frac{2rN}{K} - \bar{\beta}P$$

At fixed point $N^* = K(1-\bar{\beta}P^*/r)$, substitute to simplify.
Step 7: $\partial f/\partial P$: only $-\bar{\beta}NP$ depends on P:

$$\frac{\partial f}{\partial P} = -\bar{\beta}N$$

3.2 Building the Full Jacobian Matrix

Standard Jacobian: 2-Variable System

For $\dot{x} = f(x,y)$ and $\dot{y} = g(x,y)$:

$$J = \begin{pmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial g/\partial x & \partial g/\partial y \end{pmatrix}$$

Then evaluate at the fixed point (x^*, y^*). The resulting matrix J^* tells you stability.

Worked Example: Full Jacobian for Lotka-Volterra

$f = rN - \bar{\beta}NP$, $g = \varepsilon\bar{\beta}NP - ZP$. From above:

$$J = \begin{pmatrix} r - \bar{\beta}P & -\bar{\beta}N \\ \varepsilon\bar{\beta}P & \varepsilon\bar{\beta}N - Z \end{pmatrix}$$

Fixed point $(N^*, P^*) = (Z/\varepsilon\bar{\beta},\ r/\bar{\beta})$. At FP: $r - \bar{\beta}P^* = r - r = 0$ and $\varepsilon\bar{\beta}N^* - Z = Z - Z = 0$:

$$J^* = \begin{pmatrix} 0 & -\bar{\beta}N^* \\ \varepsilon\bar{\beta}P^* & 0 \end{pmatrix} = \begin{pmatrix} 0 & -Z/\varepsilon \\ \varepsilon r & 0 \end{pmatrix}$$

4. Separation of Variables (Solving ODEs)

The Method

To solve $\frac{dN}{dt} = f(N)\cdot g(t)$:

1. Move all N's to the left: $\frac{dN}{f(N)} = g(t)\,dt$
2. Integrate both sides: $\displaystyle\int \frac{dN}{f(N)} = \int g(t)\,dt + C$
3. Solve for $N(t)$
4. Apply initial condition to find C

4.1 Case 1: Exponential Growth $\dot{N} = rN$

Step 1: Move N to left:
$\dfrac{dN}{N} = r\,dt$
Step 2: Integrate both sides:

$$\int \frac{1}{N}\,dN = \int r\,dt \quad\Longrightarrow\quad \ln|N| = rt + C$$

Step 3: Exponentiate both sides:

$$|N| = e^{rt+C} = e^C e^{rt}$$

Step 4: Since $N > 0$, drop absolute value; rename $e^C \to N_0$:

$$\boxed{N(t) = N_0 e^{rt}}$$

Step 5: Apply initial condition: $N(0) = N_0 e^0 = N_0$. So $N_0 = N(0)$. ✓

4.2 Case 2: Power-of-Time $\dot{N} = rN/t$

Step 1: Separate:
$\frac{dN}{N} = r\frac{dt}{t}$
Step 2: Integrate:

$$\int \frac{dN}{N} = r\int \frac{dt}{t} \quad \Longrightarrow \quad \ln N = r\ln t + C$$

Step 3: Use log power rule (5) on right: $r\ln t = \ln(t^r)$:

$$\ln N = \ln(t^r) + C$$

Step 4: Exponentiate:

$$N = e^{\ln(t^r)} \cdot e^C = t^r \cdot e^C$$

Step 5: Apply IC at $t = 1$: $N(1) = 1^r \cdot e^C = e^C$, so $e^C = N(1)$:

$$\boxed{N(t) = N(1)\, t^r}$$

4.3 Case 3: Inverse-Time $\dot{N} = r/t$

Step 1: Separate: N doesn't appear on right, so just:

$$dN = \frac{r}{t}\, dt$$

Step 2: Integrate both sides:

$$\int dN = r\int \frac{dt}{t} \quad \Longrightarrow \quad N = r\ln t + C$$

Step 3: Apply IC at $t = 1$: $N(1) = r\ln 1 + C = r \cdot 0 + C = C$:

$$\boxed{N(t) = N(1) + r\ln t}$$

4.4 Case 4: Super-Exponential $\dot{N} = rN^p$ $(p \neq 1)$

Step 1: Separate:

$$\frac{dN}{N^p} = r\, dt \quad \Longrightarrow \quad N^{-p}\, dN = r\, dt$$

Step 2: Integrate left using power rule ($\int N^{-p} dN = \frac{N^{-p+1}}{-p+1}$):

$$\frac{N^{1-p}}{1-p} = rt + C$$

Step 3: Solve for N^{1-p}:

$$N^{1-p} = (1-p)(rt + C)$$

Step 4: Raise both sides to power $\frac{1}{1-p}$:

$$N(t) = [(1-p)(rt+C)]^{\frac{1}{1-p}}$$

Step 5: For $p = 2$ specifically ($\dot{N} = rN^2$):

$$\int N^{-2} dN = -N^{-1} = -\frac{1}{N} = rt + C \quad \Longrightarrow \quad N(t) = \frac{-1}{rt+C} = \frac{1}{C' - rt}$$

where $C' = -C$. At $t = 0$: $N(0) = 1/C'$, so $C' = 1/N_0$:

$$\boxed{N(t) = \frac{N_0}{1 - rN_0 t}}$$

Blows up at $t^* = 1/(rN_0)$.

4.5 Case 5: Logistic Growth $\dot{N} = rN(1 - N/K)$

Step 1: Separate:

$$\frac{dN}{N(1-N/K)} = r\,dt$$

Step 2: Partial fractions on left. Write:

$$\frac{1}{N(1-N/K)} = \frac{A}{N} + \frac{B}{1-N/K}$$

Multiply through by $N(1-N/K)$: $1 = A(1-N/K) + BN$.
At $N = 0$: $1 = A$. At $N = K$: $1 = BK$, so $B = 1/K$. Thus:

$$\frac{1}{N(1-N/K)} = \frac{1}{N} + \frac{1/K}{1-N/K}$$

Step 3: Integrate both sides:

$$\int \left(\frac{1}{N} + \frac{1/K}{1-N/K} \right) dN = rt + C$$

For the second integral, let $u = 1 - N/K$, $du = -dN/K$, so $dN/K = -du$:

$$\int \frac{1/K}{1-N/K} dN = -\int \frac{1}{u} du = -\ln|u| = -\ln(1-N/K)$$

So:

$$\ln N - \ln(1-N/K) = rt + C \quad \Longrightarrow \quad \ln \frac{N}{1-N/K} = rt + C$$

Step 4: Exponentiate:

$$\frac{N}{1-N/K} = e^{rt+C} = e^C e^{rt}$$

Step 5: Solve for N: let $e^C = A$:

$$N = Ae^{rt}(1-N/K) = Ae^{rt} - \frac{Ae^{rt}}{K} N$$

$$N\left(1 + \frac{Ae^{rt}}{K}\right) = Ae^{rt}$$

$$N(t) = \frac{Ae^{rt}}{1 + Ae^{rt}/K} = \frac{K}{K/(Ae^{rt}) + 1} = \frac{K}{1 + (K/A)e^{-rt}}$$

Step 6: At $t = 0$: $N_0 = K/(1 + K/A)$. Solving: $K/A = K/N_0 - 1 = (K - N_0)/N_0$. So:

$$\boxed{N(t) = \frac{K}{1 + \frac{K - N_0}{N_0} e^{-rt}}}$$

As $t \to \infty$: $e^{-rt} \to 0$, so $N \to K$. ✓

(!) Common Mistake

When using partial fractions, always verify your result by substituting back. Check that both terms sum to $\frac{1}{N(1-N/K)}$ before integrating.

5. Taylor Expansion — Complete Treatment

Taylor Series Formula

$$f(t+\Delta t) = f(t) + f'(t)\,\Delta t + \frac{f''(t)}{2!}(\Delta t)^2 + \frac{f'''(t)}{3!}(\Delta t)^3 + \frac{f^{(4)}(t)}{4!}(\Delta t)^4 + \cdots$$

Euler's method = keep only the first two terms.
RK4 = conceptually equivalent to keeping four terms.
Perfect Euler = all higher derivatives vanish $\Rightarrow$ infinite steps are exact.

5.1 How to Compute the 4th-Order Taylor Term

Recipe for the 4th-Order Term

Given $N(t)$:

1. Differentiate $N(t)$ four times to get $N^{(4)}(t)$.
2. Plug in your expansion point t_0.
3. The 4th-order term is $\dfrac{N^{(4)}(t_0)}{4!}(\Delta t)^4 = \dfrac{N^{(4)}(t_0)}{24}(\Delta t)^4$.

Worked Example: $N(t) = N(1)t^r$ with $r = 10$

Step 1: First derivative:

$$N'(t) = N(1)\cdot r\cdot t^{r-1} = 10\,N(1)\,t^9$$

Step 2: Second derivative:

$$N''(t) = 10\cdot 9\,N(1)\,t^8 = 90\,N(1)\,t^8$$

Step 3: Third derivative:

$$N'''(t) = 90\cdot 8\,N(1)\,t^7 = 720\,N(1)\,t^7$$

Step 4: Fourth derivative:

$$N^{(4)}(t) = 720\cdot 7\,N(1)\,t^6 = 5040\,N(1)\,t^6$$

Step 5: 4th-order Taylor term (evaluated at $t_0 = 1$):

$$\frac{N^{(4)}(1)}{4!}(\Delta t)^4 = \frac{5040\,N(1)\cdot 1}{24}(\Delta t)^4 = \boxed{210\,N(1)\,(\Delta t)^4}$$

Note: $r(r-1)(r-2)(r-3) = 10\cdot 9\cdot 8\cdot 7 = 5040$ *and* $5040/24 = 210$.

Worked Example: $N(t) = N(1) + r\ln t$

Step 1: First derivative: $N' = r/t = rt^{-1}$
Step 2: Second derivative: $N'' = -rt^{-2} = -r/t^2$
Step 3: Third derivative: $N''' = 2rt^{-3} = 2r/t^3$
Step 4: Fourth derivative: $N^{(4)} = -6rt^{-4} = -6r/t^4$

At $t_0 = 1$:

$$\text{4th-order term} = \frac{-6r}{4!}(\Delta t)^4 = \frac{-6r}{24}(\Delta t)^4 = -\frac{r}{4}(\Delta t)^4$$

5.2 Key Taylor Expansions for Early/Late Time Limits

Expansions to Memorize

For $x \ll 1$ (early times / small quantities):

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots \tag{9}$$

$$\ln(1+x) \approx x - \frac{x^2}{2} + \cdots \approx x \quad (x \ll 1) \tag{10}$$

$$(1+x)^n \approx 1 + nx \quad (x \ll 1) \tag{11}$$

$$\frac{1}{1+x} \approx 1 - x \quad (x \ll 1) \tag{12}$$

Leading-order simplification strategy (exam skill):

- If $x(t) \ll 1$: replace x^n by $x^{\text{lowest power}}$ everywhere.
- If $x(t) \gg 1$: drop the lower-power terms, keep the dominant one.

Worked Example: 2025 Exam-Style: Simplify at early times ($x \ll 1$, $t \ll \tau$)

Growth rate:

$$\frac{dx}{dt} = r\frac{(c_7x^7t + c_5x^5t + c_3x^3t)}{(a_8x^8t + a_4x^4t + a_2x^2t)} + \cdots$$

When $x \ll 1$, the lowest power dominates in each polynomial. Numerator: lowest power of x is x^3 (from c_3x^3t). Denominator: lowest power is x^2 (from a_2x^2t). Keep only those terms:

$$\frac{dx}{dt} \approx r\frac{c_3x^3t}{a_2x^2t} = \frac{rc_3}{a_2}x$$

Define $r_{\text{eff}} = rc_3/a_2$. This is exponential: $\dot{x} = r_{\text{eff}}x$.
At late times ($x \gg 1$): highest power dominates. Numerator: c_7x^7t. Denominator: a_8x^8t:

$$\frac{dx}{dt} \approx r\frac{c_7x^7t}{a_8x^8t} = \frac{rc_7}{a_8}\cdot\frac{1}{x}\cdot\frac{1}{1} \quad \Longrightarrow \quad \frac{dx}{dt} \approx \frac{rc_7}{a_8x}$$

This gives $x\,dx = \frac{rc_7}{a_8}dt \quad \Longrightarrow \quad x^2 = \frac{2rc_7}{a_8}t + C \quad \Longrightarrow \quad x(t) \propto \sqrt{t}$.

5.3 Finding the "Perfect Euler" Space

The Strategy

Euler's method is **exact** when all higher derivatives of N (or the transformed variable) vanish. This happens when N (after transformation) is **linear** in time.
Procedure:

1. Guess a transformed variable $u = h(N)$ and new time $s = q(t)$.
2. Compute du/ds.
3. If $du/ds =$ constant, you found the linear (Euler-perfect) space.

Worked Example: Exponential growth $\dot{N} = rN$

Try $u = \ln N$, $s = t$:

$$\frac{du}{ds} = \frac{d\ln N}{dt} = \frac{1}{N}\frac{dN}{dt} = \frac{1}{N}(rN) = r = \text{const}$$

Space: $\ln N$ **vs** t**.** Euler is exact. All 2nd and higher derivatives of u w.r.t. s are zero.

Worked Example: Power-of-time $\dot{N} = rN/t$

Try $u = \ln N$, $s = \ln t$:

$$\frac{du}{ds} = \frac{d\ln N}{d\ln t} = \frac{d\ln N/dt}{d\ln t/dt} = \frac{\dot{N}/N}{1/t} = \frac{r/t}{1/t} = r = \text{const}$$

Space: $\ln N$ **vs** $\ln t$ **(log-log).**

Worked Example: Inverse-time $\dot{N} = r/t$

Try $u = N$, $s = \ln t$:

$$\frac{du}{ds} = \frac{dN}{d\ln t} = \frac{dN/dt}{d\ln t/dt} = \frac{r/t}{1/t} = r = \text{const}$$

Space: N **vs** $\ln t$ **(semi-log in** t**).**

Worked Example: Super-exponential $\dot{N} = rN^2$

Try $u = 1/N = N^{-1}$, $s = t$:

$$\frac{du}{ds} = \frac{d(1/N)}{dt} = -\frac{1}{N^2}\frac{dN}{dt} = -\frac{1}{N^2}(rN^2) = -r = \text{const}$$

Space: $1/N$ **vs** t**.**

6. Eigenvalue Analysis — Full Treatment

6.1 Finding Fixed Points

Fixed Point Procedure

A fixed point $\vec{X}^*$ satisfies $d\vec{X}/dt = 0$ simultaneously for all equations.

1. Set each equation equal to zero.
2. Factor — **don't divide by a variable** (you'll lose the $X^* = 0$ solution).
3. Enumerate all combinations.

Worked Example: Lotka-Volterra fixed points

$$\dot{N} = N(r - \bar{\beta}P) = 0 \quad \Longrightarrow \quad N = 0 \quad \text{or} \quad P = r/\bar{\beta}$$
$$\dot{P} = P(\varepsilon\bar{\beta}N - Z) = 0 \quad \Longrightarrow \quad P = 0 \quad \text{or} \quad N = Z/(\varepsilon\bar{\beta})$$

Combinations: $(N^* = 0, P^* = 0)$ trivial; $(N^* = Z/(\varepsilon\bar{\beta}),\ P^* = r/\bar{\beta})$ coexistence.

6.2 The 2×2 Eigenvalue Formula — Every Step

From Tr and Det to Eigenvalues

For any 2×2 matrix M:

$$\text{Tr}(M) = M_{11} + M_{22}, \qquad \text{Det}(M) = M_{11}M_{22} - M_{12}M_{21}$$

The characteristic equation is:

$$\lambda^2 - \text{Tr} \cdot \lambda + \text{Det} = 0$$

Quadratic formula:

$$\lambda_\pm = \frac{\text{Tr} \pm \sqrt{\text{Tr}^2 - 4\,\text{Det}}}{2}$$

Equivalently:

$$\lambda_\pm = \frac{\text{Tr}}{2}\left(1 \pm \sqrt{1 - \frac{4\,\text{Det}}{\text{Tr}^2}}\right)$$

Worked Example: Full eigenvalue calculation for logistic LV

$$A = \begin{pmatrix} -r/K & -\bar{\beta} \\ \varepsilon\bar{\beta} & 0 \end{pmatrix}$$

Step 5: Compute Tr and Det:

$$\text{Tr} = -r/K + 0 = -r/K$$
$$\text{Det} = (-r/K)(0) - (-\bar{\beta})(\varepsilon\bar{\beta}) = 0 + \varepsilon\bar{\beta}^2 = \varepsilon\bar{\beta}^2$$

Step 6: Discriminant $\Delta = \text{Tr}^2 - 4\,\text{Det}$:

$$\Delta = \frac{r^2}{K^2} - 4\varepsilon\bar{\beta}^2$$

Step 7: Eigenvalues:

$$\lambda_\pm = \frac{-r/K \pm \sqrt{r^2/K^2 - 4\varepsilon\bar{\beta}^2}}{2}$$

Step 8: Both have negative real part since Tr $= -r/K < 0$ and Det $= \varepsilon\bar{\beta}^2 > 0$. Stable.

If $\Delta < 0$ (i.e. $4\varepsilon\bar{\beta}^2 > r^2/K^2$): complex eigenvalues $\Rightarrow$ **stable spiral** (oscillates into equilibrium). If $\Delta > 0$: **stable node** (no oscillations).

6.3 Stability Classification Table

Decision Table for 2×2 Systems

Tr	Det	Discriminant	Classification
< 0	> 0	> 0	Stable node (real $\lambda < 0$)
< 0	> 0	< 0	Stable spiral (complex λ, Re < 0)
$= 0$	> 0	< 0	Neutral oscillations (pure imaginary λ)
> 0	> 0	any	Unstable (node or spiral)
any	< 0	> 0	Saddle point (always unstable)
> 0	> 0	< 0	Unstable spiral

6.4 Interaction Matrix A vs. Standard Jacobian J^*

When to Use Each

Interaction matrix A:

- Write each ODE as: $\dot{X}_i/X_i = \sum_j A_{ij}X_j + \text{constants}$
- A_{ij} = coefficient of X_j in $\dot{X}_i/X_i$
- **Only valid at non-trivial fixed points** ($X_i^* \neq 0$ for all i)
- **Cannot use** when interactions are nonlinear (Type II) or when any FP component is zero

Standard Jacobian J^*:

- Always valid
- Must compute 4 partial derivatives, evaluate at FP
- Required when FP has zero components (disease-free state, trivial predator state)

Worked Example: Building the A matrix for LV

$\dot{N}/N = r - \bar{\beta}P \Rightarrow$ coefficients: $A_{NN} = 0$ (no N term), $A_{NP} = -\bar{\beta}$.
$\dot{P}/P = \varepsilon\bar{\beta}N - Z \Rightarrow$ coefficients: $A_{PN} = \varepsilon\bar{\beta}$, $A_{PP} = 0$.

$$A = \begin{pmatrix} A_{NN} & A_{NP} \\ A_{PN} & A_{PP} \end{pmatrix} = \begin{pmatrix} 0 & -\bar{\beta} \\ \varepsilon\bar{\beta} & 0 \end{pmatrix}$$

Note: A is **not** the same as J^* but gives same eigenvalue signs at non-trivial FP.

6.5 Complex Eigenvalues — What They Mean

Interpreting $\lambda = a \pm i\omega$

If the discriminant $\Delta < 0$:

$$\lambda_\pm = \frac{\text{Tr}}{2} \pm i\frac{\sqrt{|\Delta|}}{2} = \underbrace{\frac{\text{Tr}}{2}}_{a} \pm i\underbrace{\frac{\sqrt{4\text{Det} - \text{Tr}^2}}{2}}_{\omega}$$

The solution to $\dot{\delta x} = J^* \delta x$ is then:

$$\delta x(t) \propto e^{at} \cos(\omega t + \phi)$$

- $a = \mathrm{Tr}/2$: controls amplitude. If $a < 0$: decays (stable spiral). If $a > 0$: grows (unstable spiral).
- ω: oscillation frequency. Larger ω = faster oscillations.
- $a = 0$ (i.e. $\mathrm{Tr} = 0$): **neutral limit cycles** (LV standard case).

7. Basic Integrals Reference

Every Integral You Need

Integral	Result
$\int x^n \, dx$	$\frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
$\int \frac{1}{x} \, dx$	$\ln\lvert x\rvert + C$
$\int e^{rx} \, dx$	$\frac{1}{r} e^{rx} + C$
$\int e^x \, dx$	$e^x + C$
$\int \frac{1}{a+bx} \, dx$	$\frac{1}{b} \ln\lvert a+bx\rvert + C$
$\int \frac{1}{1-u} \, du$	$-\ln\lvert 1-u\rvert + C$
$\int c \, dx$	$cx + C$
$\int 0 \, dx$	C

Substitution rule: If the integral has the form $\int f(g(x))g'(x) \, dx$, let $u = g(x)$, $du = g'(x)dx$, and integrate $\int f(u) \, du$.

Worked Example: Substitution: $\int \frac{1/K}{1-N/K} \, dN$

Step 1: Let $u = 1 - N/K$. Then $du = -dN/K$, so $dN/K = -du$, meaning $dN = -K \, du$.
Step 2: Rewrite integrand:

$$\int \frac{1/K}{u} \cdot (-K \, du) = -\int \frac{1}{u} \, du$$

Step 3: Integrate:

$$= -\ln\lvert u\rvert + C = -\ln\left\lvert 1 - \frac{N}{K}\right\rvert + C$$

8. QSSA (Quasi-Steady-State Approximation) — Steps

QSSA Procedure

Used for Michaelis-Menten and analogous systems.

1. Write all ODEs.
2. Identify the fast/intermediate variable (usually the complex $[SE]$ or analogously I in SIRS).
3. Set its derivative to zero: $d[\text{complex}]/dt = 0$.
4. Use the conservation law to eliminate one variable.
5. Solve for complex in terms of substrate and total enzyme.
6. Substitute to get the production rate.

Worked Example: Standard Michaelis-Menten: Full Steps

Reaction: $S + E \rightleftharpoons [SE] \to P + E$.
Equations:

$$\begin{aligned}\dot{S} &= -k_1 SE + k_{-1}[SE] \\ \dot{[SE]} &= k_1 SE - k_{-1}[SE] - k_2[SE] \\ \dot{P} &= k_2[SE] \\ \dot{E} &= -k_1 SE + (k_{-1} + k_2)[SE]\end{aligned}$$

Step 1: Conservation: $\dot{E} + \dot{[SE]} = 0 \Rightarrow E_0 = E + [SE] =$ **const. So** $E = E_0 - [SE]$.

Step 2: Apply QSSA: set $\dot{[SE]} = 0$:

$$k_1 SE - (k_{-1} + k_2)[SE] = 0$$

Step 3: Substitute $E = E_0 - [SE]$:

$$k_1 S(E_0 - [SE]) - (k_{-1} + k_2)[SE] = 0$$

Step 4: Expand and collect $[SE]$ **terms:**

$$k_1 SE_0 = [SE]\,(k_1 S + k_{-1} + k_2)$$

Step 5: Solve for $[SE]$:

$$[SE] = \frac{k_1 SE_0}{k_1 S + k_{-1} + k_2}$$

Step 6: Divide numerator and denominator by k_1:

$$[SE] = \frac{SE_0}{S + \dfrac{k_{-1} + k_2}{k_1}} = \frac{SE_0}{S + K_M}, \quad K_M = \frac{k_{-1} + k_2}{k_1}$$

Step 7: Production rate:

$$\frac{dP}{dt} = k_2[SE] = \frac{k_2 E_0 S}{K_M + S} = \frac{V_{\max} S}{K_M + S}, \quad V_{\max} = k_2 E_0$$

Step 8: Limiting cases:

$S \ll K_M$: $\dot{P} \approx \frac{V_{\max}}{K_M} S$ (linear). $S \gg K_M$: $\dot{P} \approx V_{\max}$ (constant).

9. R_0 and SIR Stability — Complete Steps

Worked Example: Standard SIR: Eigenvalue Calculation at Disease-Free FP

Equations: $\dot{S} = -\beta SI$, $\dot{I} = \beta SI - \gamma I$, $\dot{R} = \gamma I$.
$N = S + I + R$ conserved. Work with S and I only ($R = N - S - I$).
Step 1: Disease-free FP: $I^* = 0$, $S^* = N$.

Step 2: Build Jacobian (2×2 for S **and** I**):**

$$J = \begin{pmatrix} \partial(-\beta SI)/\partial S & \partial(-\beta SI)/\partial I \\ \partial(\beta SI - \gamma I)/\partial S & \partial(\beta SI - \gamma I)/\partial I \end{pmatrix} = \begin{pmatrix} -\beta I & -\beta S \\ \beta I & \beta S - \gamma \end{pmatrix}$$

Step 3: Evaluate at $(S^* = N, I^* = 0)$:

$$J^* = \begin{pmatrix} 0 & -\beta N \\ 0 & \beta N - \gamma \end{pmatrix}$$

Step 4: This is upper triangular. Eigenvalues of triangular matrices are the diagonal entries:

$$\lambda_1 = 0, \quad \lambda_2 = \beta N - \gamma$$

Step 5: Disease spreads iff $\lambda_2 > 0$:

$$\beta N - \gamma > 0 \quad \Longrightarrow \quad \frac{\beta N}{\gamma} > 1 \quad \Longrightarrow \quad \boxed{R_0 = \frac{\beta N}{\gamma} > 1}$$

Reading Eigenvalues from Triangular Matrices

If J^* is upper or lower triangular, the eigenvalues are just the diagonal entries. No quadratic needed.

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \Rightarrow \lambda_1 = a,\ \lambda_2 = d \qquad \begin{pmatrix} a & 0 \\ c & d \end{pmatrix} \Rightarrow \lambda_1 = a,\ \lambda_2 = d$$

This is the most common shortcut on CaSB 150 disease-model eigenvalue problems.

10. Quick Reference: Algebra of Inequalities

Sign Rules for Inequalities

1. Adding/subtracting a number: $a > b \Rightarrow a + c > b + c$ (direction preserved)
2. Multiplying/dividing by a **positive** number: direction preserved.
3. Multiplying/dividing by a **negative** number: **direction flips**.
4. $a > b$ and $c > 0$: $ac > bc$.
5. $a > b$ and $c < 0$: $ac < bc$.

Worked Example: Solving $\lambda_2 = \varepsilon\bar{\beta}N - Z > 0$ for condition on N

$$\varepsilon\bar{\beta}N - Z > 0 \quad \Longrightarrow \quad \varepsilon\bar{\beta}N > Z \quad \Longrightarrow \quad N > \frac{Z}{\varepsilon\bar{\beta}}$$

All divisions by $\varepsilon, \bar{\beta} > 0$, so direction preserved. Biological meaning: prey must exceed the threshold $N^* = Z/(\varepsilon\bar{\beta})$ for predator to grow.

(!) Common Mistake

When solving $-rN/K > 0$, dividing by $-r/K < 0$ **flips** the inequality:

$$-\frac{r}{K}N > 0 \quad \Longrightarrow \quad N < 0$$

This correctly identifies that logistic self-interaction is only stabilizing for $N > 0$ (no biologically negative populations).

11. Putting It All Together: Exam Workflow

Workflow for Any Exam Problem

Growth equation problem:

1. Simplify using early/late time limits (keep lowest/highest power of small/large quantity).
2. Separate variables and integrate to solve.
3. Find the linear space: compute du/ds for candidate $u = h(N)$, $s = q(t)$; check if constant.
4. Compute 4th-order Taylor term: differentiate 4 times, divide by $4! = 24$.

Predator-prey / SIR problem:

1. Check conservation (sum all ODEs; if $= 0$, eliminate one variable).
2. Find fixed points by setting each ODE to zero and factoring.
3. Decide: Jacobian or A matrix? (Any zero FP component $\Rightarrow$ Jacobian only.)
4. Compute Tr and Det.
5. Get eigenvalues; classify stability.
6. State biological conclusion.

Michaelis-Menten / QSSA problem:

1. Write all ODEs.
2. Find conservation laws (set $\dot{E} + \dot{[SE]} = 0$ etc.).
3. Set $\dot{[SE]} = 0$.
4. Substitute conservation to eliminate E.
5. Solve algebraically for $[SE]$.
6. Compute $\dot{P} = k_2[SE]$.